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JEE Main Physics Practice Sheet — DPP-050

Topic: Basic Concepts of Rotational Motion: Angular Displacement, Velocity & Acceleration**Time Allowed:** 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** The angular displacement of a rotating wheel varies with time as $\theta(t) = 2t^3 - 6t^2 + 10$ rad. The angular acceleration of the wheel becomes zero at time t equal to:
- (A) 1 s
(B) 2 s
(C) 0.5 s
(D) 3 s
- Q2.** A wheel starts rotating from rest with a uniform angular acceleration of 4 rad/s^2 . The total angle turned by the wheel in the first 5 seconds is:
- (A) 25 rad
(B) 100 rad
(C) 50 rad
(D) 20 rad
- Q3.** Which of the following quantities in rotational kinematics is an axial vector?
- (A) Large finite angular displacement
(B) Moment of inertia
(C) Rotational kinetic energy
(D) Angular velocity
- Q4.** A flywheel rotating at 1200 rpm slows down uniformly to 600 rpm in 10 seconds. The number of revolutions completed by the flywheel during this time interval is:
- (A) 150
(B) 200
(C) 100
(D) 300
- Q5.** The position vector of a particle rotating in a circle of radius $r = 2 \text{ m}$ is $\vec{r} = (2\hat{i}) \text{ m}$ and its angular velocity vector is $\vec{\omega} = (3\hat{k}) \text{ rad/s}$. The linear velocity vector $\vec{v} = \vec{\omega} \times \vec{r}$ of the particle is:
- (A) $-6\hat{j} \text{ m/s}$
(B) $6\hat{j} \text{ m/s}$
(C) $6\hat{i} \text{ m/s}$
(D) $6\hat{k} \text{ m/s}$
- Q6.** A rigid body rotates with an angular velocity $\omega(t) = (4t - 3) \text{ rad/s}$. The total angular displacement between $t = 1 \text{ s}$ and $t = 3 \text{ s}$ is:
- (A) 10 rad
(B) 8 rad
(C) 12 rad
(D) 6 rad
- Q7.** For a particle executing circular motion of radius R with angular speed ω and angular acceleration α , the magnitude of the net linear acceleration is:
- (A) $R\sqrt{\omega^2 + \alpha^2}$
(B) $R\sqrt{\omega^4 + \alpha^2}$
(C) $R(\omega^2 + \alpha)$
(D) $R\alpha$
- Q8.** A motor rotates at a constant speed of $\omega = 30 \text{ rad/s}$. When switched off, it comes to rest after completing 45 radians under constant angular retardation. The magnitude of the angular retardation is:
- (A) 5 rad/s^2
(B) 20 rad/s^2
(C) 15 rad/s^2
(D) 10 rad/s^2
- Q9.** A particle moves along a circle of radius $R = 0.5 \text{ m}$ such that its speed increases at a constant rate $a_t = 3 \text{ m/s}^2$. The angular acceleration α of the particle is:
- (A) 1.5 rad/s^2
(B) 12 rad/s^2
(C) 6 rad/s^2
(D) 0.75 rad/s^2
- Q10.** If the angular acceleration of a wheel is related to its angular velocity as $\alpha = -k\omega$ (where k is a positive constant), the angular velocity after time t from initial speed ω_0 is:
- (A) $\omega_0(1 - kt)$
(B) $\omega_0 e^{-kt}$
(C) $\frac{\omega_0}{1 + kt}$
(D) $\omega_0 e^{-kt^2/2}$
- Q11.** A ceiling fan is rotating at 120 rpm. When the power is turned off, it completes 20 revolutions before coming to a stop with uniform deceleration. The time taken to stop is:
- (A) 20 s
(B) 10 s
(C) 15 s
(D) 30 s
- Q12.** The angular acceleration of a body rotating about a fixed axis is $\alpha = 3t^2 \text{ rad/s}^2$. If it starts from rest at $t = 0$, its angular displacement at $t = 2 \text{ s}$ is:
- (A) 8 rad

- (B) 12 rad
(C) 4 rad
(D) 16 rad
- Q13.** A particle moves in a circle of radius $r = 1$ m. Starting from rest, it is accelerated at a constant angular rate $\alpha = 2 \text{ rad/s}^2$. The time at which the magnitude of radial acceleration equals the magnitude of tangential acceleration is:
- (A) $\frac{1}{\sqrt{2}}$ s
(B) 1 s
(C) $\sqrt{2}$ s
(D) 2 s
- Q14.** The ratio of the angular speed of the second hand to that of the hour hand of a standard mechanical clock is:
- (A) 60 : 1
(B) 120 : 1
(C) 360 : 1
(D) 720 : 1
- Q15.** A wheel rotates with an angular speed $\omega = (2\theta + 5) \text{ rad/s}$, where θ is in radians. The angular acceleration α when $\theta = 5 \text{ rad}$ is:
- (A) 15 rad/s^2
(B) 60 rad/s^2
(C) 30 rad/s^2
(D) 45 rad/s^2
- Q16.** Two gears A and B are in mesh. The radius of gear A is $r_A = 10 \text{ cm}$ and that of gear B is $r_B = 25 \text{ cm}$. If gear A rotates with an angular speed of 50 rad/s , the angular speed of gear B is:
- (A) 20 rad/s
(B) 125 rad/s
(C) 25 rad/s
(D) 10 rad/s
- Q17.** An infinitesimal angular displacement $d\vec{\theta}$ is treated as a true vector because:
- (A) It has both magnitude and direction
(B) It obeys the commutative law of vector addition ($d\vec{\theta}_1 + d\vec{\theta}_2 = d\vec{\theta}_2 + d\vec{\theta}_1$)
(C) It is always perpendicular to the linear velocity
(D) Its magnitude is measured in radians
- Q18.** A wheel starts from rest and accelerates uniformly with $\alpha = 4 \text{ rad/s}^2$. The angular displacement during the 3rd second of its motion is:
- (A) 18 rad
(B) 12 rad
(C) 10 rad
(D) 14 rad
- Q19.** A point P on the rim of a rotating disc of radius $R = 0.2 \text{ m}$ has a linear speed of $v = 4 \text{ m/s}$ and net linear acceleration of $a = 80\sqrt{2} \text{ m/s}^2$. The angular acceleration α of the disc is:
- (A) 400 rad/s^2
(B) 200 rad/s^2
(C) 800 rad/s^2
(D) 100 rad/s^2
- Q20.** A shaft rotating at angular speed ω_0 is subjected to a retarding angular acceleration $\alpha = -c\sqrt{\omega}$. The time required for the shaft to come to complete rest is:
- (A) $\frac{\sqrt{\omega_0}}{c}$
(B) $\frac{2\sqrt{\omega_0}}{c}$
(C) $\frac{2\omega_0}{c}$
(D) $\frac{\omega_0^2}{2c}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-050)

Basic Concepts of Rotational Motion: Angular Displacement, Velocity & Acceleration

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	A	16	A
2	C	7	B	12	C	17	B
3	D	8	D	13	A	18	C
4	A	9	C	14	D	19	A
5	B	10	B	15	C	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)

Given $\theta(t) = 2t^3 - 6t^2 + 10$:

$$\omega(t) = \frac{d\theta}{dt} = 6t^2 - 12t$$

$$\alpha(t) = \frac{d\omega}{dt} = 12t - 12$$

Setting $\alpha(t) = 0 \Rightarrow 12t - 12 = 0 \Rightarrow t = 1$ s.

Sol 2. Option (C)

Using rotational kinematic equation:

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

With $\omega_0 = 0$, $\alpha = 4$ rad/s², $t = 5$ s:

$$\theta = 0 + \frac{1}{2}(4)(5^2) = 2 \times 25 = 50 \text{ rad}$$

Sol 3. Option (D)

Angular velocity $\vec{\omega}$ and infinitesimal angular displacement $d\vec{\theta}$ are axial vectors whose direction is given by the right-hand thumb rule along the axis of rotation. Finite angular displacements do not commute under addition, so they are not vectors.

Sol 4. Option (A)

Initial frequency: $N_1 = 1200$ rpm = 20 rps.

Final frequency: $N_2 = 600$ rpm = 10 rps.

Number of revolutions in $t = 10$ s:

$$n = \left(\frac{N_1 + N_2}{2} \right) t = \left(\frac{20 + 10}{2} \right) (10) = 15 \times 10 = 150$$

Sol 5. Option (B)

Linear velocity is given by the cross product:

$$\vec{v} = \vec{\omega} \times \vec{r} = (3\hat{k}) \times (2\hat{i}) = 6(\hat{k} \times \hat{i}) = 6\hat{j} \text{ m/s}$$

Sol 6. Option (A)

Angular displacement:

$$\Delta\theta = \int_1^3 \omega(t) dt = \int_1^3 (4t - 3) dt$$

$$\Delta\theta = [2t^2 - 3t]_1^3 = [2(9) - 3(3)] - [2(1) - 3(1)]$$

$$\Delta\theta = [18 - 9] - [2 - 3] = 9 - (-1) = 10 \text{ rad}$$

Sol 7. Option (B)

The radial (centripetal) acceleration is $a_r = R\omega^2$.

The tangential acceleration is $a_t = R\alpha$.

Net linear acceleration:

$$a = \sqrt{a_r^2 + a_t^2} = \sqrt{(R\omega^2)^2 + (R\alpha)^2} = R\sqrt{\omega^4 + \alpha^2}$$

Sol 8. Option (D)

Using $\omega^2 = \omega_0^2 - 2\alpha\theta$:

$$0 = 30^2 - 2\alpha(45) \Rightarrow 90\alpha = 900 \Rightarrow \alpha = 10 \text{ rad/s}^2$$

Sol 9. Option (C)

Tangential acceleration and angular acceleration are related by:

$$a_t = R\alpha \Rightarrow \alpha = \frac{a_t}{R} = \frac{3 \text{ m/s}^2}{0.5 \text{ m}} = 6 \text{ rad/s}^2$$

Sol 10. Option (B)

Given $\frac{d\omega}{dt} = -k\omega$:

$$\frac{d\omega}{\omega} = -k dt \Rightarrow \int_{\omega_0}^{\omega} \frac{d\omega}{\omega} = -k \int_0^t dt$$

$$\ln\left(\frac{\omega}{\omega_0}\right) = -kt \Rightarrow \omega(t) = \omega_0 e^{-kt}$$

Sol 11. Option (A)

Initial speed $\omega_0 = 120$ rpm = $120 \times \frac{2\pi}{60} = 4\pi$ rad/s.

Total angle $\theta = 20$ rev = $20 \times 2\pi = 40\pi$ rad.

Using $\theta = \left(\frac{\omega_0 + \omega}{2} \right) t$:

$$40\pi = \left(\frac{4\pi + 0}{2} \right) t = 2\pi t \Rightarrow t = 20 \text{ s}$$

Sol 12. Option (C)

Given $\alpha(t) = 3t^2$:

$$\omega(t) = \int_0^t 3t^2 dt = t^3$$

$$\theta(t) = \int_0^t t^3 dt = \frac{t^4}{4}$$

At $t = 2$ s:

$$\theta(2) = \frac{2^4}{4} = \frac{16}{4} = 4 \text{ rad}$$

Sol 13. Option (A)

With $\omega_0 = 0$, we have $\omega(t) = \alpha t = 2t$.

Radial acceleration: $a_r = r\omega^2 = 1(2t)^2 = 4t^2$.

Tangential acceleration: $a_t = r\alpha = 1(2) = 2$.

Setting $a_r = a_t$:

$$4t^2 = 2 \implies t^2 = \frac{1}{2} \implies t = \frac{1}{\sqrt{2}} \text{ s}$$

Sol 14. Option (D)

For second hand: period $T_s = 60 \text{ s} \implies \omega_s = \frac{2\pi}{60} \text{ rad/s}$.

For hour hand: period $T_h = 12 \text{ h} = 12 \times 3600 \text{ s} = 43200 \text{ s}$.

$$\frac{\omega_s}{\omega_h} = \frac{T_h}{T_s} = \frac{43200}{60} = 720 : 1$$

Sol 15. Option (C)

Angular acceleration in terms of θ :

$$\alpha = \omega \frac{d\omega}{d\theta}$$

Given $\omega = 2\theta + 5 \implies \frac{d\omega}{d\theta} = 2$.

At $\theta = 5 \text{ rad}$: $\omega = 2(5) + 5 = 15 \text{ rad/s}$.

$$\alpha = (15)(2) = 30 \text{ rad/s}^2$$

Sol 16. Option (A)

At the pitch point of meshed gears, tangential contact speeds are equal:

$$v_A = v_B \implies r_A \omega_A = r_B \omega_B$$

$$\omega_B = \frac{r_A \omega_A}{r_B} = \frac{10 \times 50}{25} = 20 \text{ rad/s}$$

Sol 17. Option (B)

Finite angular displacements do not commute under addition ($\theta_1 + \theta_2 \neq \theta_2 + \theta_1$), whereas infinitesimal angular displacements obey vector commutativity ($d\vec{\theta}_1 + d\vec{\theta}_2 =$

$d\vec{\theta}_2 + d\vec{\theta}_1$) and hence follow the parallelogram law of vector addition.

Sol 18. Option (C)

Angular displacement in the n -th second:

$$\theta_n = \omega_0 + \frac{1}{2}\alpha(2n - 1)$$

For $n = 3$, $\omega_0 = 0$, $\alpha = 4 \text{ rad/s}^2$:

$$\theta_3 = 0 + \frac{1}{2}(4)(2(3) - 1) = 2(5) = 10 \text{ rad}$$

Sol 19. Option (A)

Centripetal acceleration:

$$a_r = \frac{v^2}{R} = \frac{4^2}{0.2} = \frac{16}{0.2} = 80 \text{ m/s}^2$$

Given total acceleration $a = 80\sqrt{2} \text{ m/s}^2$:

$$a = \sqrt{a_r^2 + a_t^2} \implies 80\sqrt{2} = \sqrt{80^2 + a_t^2} \implies a_t = 80 \text{ m/s}^2$$

Angular acceleration:

$$\alpha = \frac{a_t}{R} = \frac{80}{0.2} = 400 \text{ rad/s}^2$$

Sol 20. Option (B)

Given $\frac{d\omega}{dt} = -c\sqrt{\omega}$:

$$\frac{d\omega}{\sqrt{\omega}} = -c dt \implies \int_{\omega_0}^0 \omega^{-1/2} d\omega = -c \int_0^t dt$$

$$[2\sqrt{\omega}]_{\omega_0}^0 = -ct \implies -2\sqrt{\omega_0} = -ct \implies t = \frac{2\sqrt{\omega_0}}{c}$$